Principles of Diet

A high calorie, high protein, high vitamin, minerals, and high fluid diet is given.

Dietary Management

Energy: High calorie diet is prescribed. Satisfactory weight can be maintained with 2500 to 3000 calories. 45-50 kcal/ kg body weight for adults and 90 kcal/kg body weight for children are suggested.

Protein: The daily protein intake of 80 to 120g or 1.5 - 2 g/kg body weight for adults is suggested.

Minerals: Calcium should be provided liberally since it is also essential for healing tuberculosis lesions. At least one litre of milk should be taken daily. The iron needs may also be increased if there has been hemorrhage.

Vitamins: The metabolism of vitamin A is adversely affected in tuberculosis. The diet should provide vitamin A. The weekly inclusion of vitamin A rich food and dietary supplementation with vitamin A is essential. Ascorbic acid intake should be increased. Large amounts of citrus fruits or supplementation of ascorbic acid is essential. Include plenty of antioxidants in the diet.

Dietary Guidelines

Easily digestible and good quality diet reduces strain on the body. During the acute stage a high-calorie diet is prescribed. One litre of milk and 3 to 4 eggs per day are given. Two or three servings of different type of fruits are suggested to meet micronutrient and antioxidant requirement.

1.3 Diet in Diseases of the Gastrointestinal Tract

Introduction

The **gastrointestinal tract** starts from the oral cavity, where food enters the mouth, continuing through the pharynx, oesophagus, stomach and intestines to the rectum and anus, where food is expelled. A**ccessory organs** such as liver and pancreas, salivary glands and the biliary tract secrete enzymes and bile to help break down food into is component nutrients.

Basic Structure

The gastrointestinal tract is a muscular tube lined by a special layer of cells, called epithelium.

i. Oral cavity

In the oral cavity or mouth the food is mechanical broken down by chewing and chopping actions of the teeth. The tongue plays an important role in chewing, swallowing, and helps the food bolus to come in contact with the teeth. It is also the sensing organ of the mouth for touch, temperature and taste using papillae.

ii. Salivary glands

Salivary glands secrete saliva that lubricate and protect both the soft and hard tissues of the oral cavity. Three pairs of salivary glands include Parotids, Submandibular and Sublingual.

Functions of Saliva

- It keeps the mouth moist and helps in speech.
- It helps in the process of mastication of the food stuff and in preparing it into a bolus suitable for digestion.

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